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Coworker Exchange: Relationships Between Coworkers, Leader–Member Exchange, and Work Attitudes

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The study of leadership exchanges is extended by studying both leader–member exchanges (LMXs) and coworker exchanges (CWXs). Data from 110 coworker dyads were used to examine relationships between LMXs and CWXs and between exchange relationships and work attitudes. As predicted, the interaction between 2 coworkers' LMX scores predicted CWX quality for the coworker dyad. Also, after controlling for LMX, greater diversity in a worker's CWX relationships was negatively related to his or her organizational commitment but not job satisfaction. The quality of a worker's CWX relationships, however, did not moderate the relationship between CWX diversity and work attitudes.

Research on leader–member exchanges (LMXs) has focused almost exclusively on the nature of the relationship between leader and follower (Gerstner & Day, 1997; Graen & Uhl-Bien, 1995; Sparrowe & Liden, 1997). Nevertheless, Graen and Uhl-Bien (1995) suggested that understanding coworker exchanges (CWXs) may be an important part of understanding how leadership processes work. CWXs also have been suggested to be an alternative influence on followers' work attitudes and performance (Seers, 1989). Moreover, it has been proposed that leadership exchanges may affect CWXs or vice versa (Graen & Uhl-Bien, 1995). For example, in-group and out-group members may have difficulty working together and may experience conflict, undermining team processes (Yukl & Van Fleet, 1990). Thus, several theorists have pointed to the value of investigating relationships among organizational members other than the leader and follower (Sparrowe & Liden, 1997; Uhl-Bien, Graen, & Scandura, 2000).

CWXs—exchanges among coworkers who report to the same supervisor—have been largely ignored in empirical research. Seers and colleagues (Seers, 1989; Seers, Petty, & Cashman, 1995) empirically examined teammember exchange (TMX), and Dunegan, Tierney, and Duchon (1992) studied work group exchange (WGX). In these works, TMX and WGX were found to be positively associated with work attitudes, perceptions of climate, efficiency, or performance. Both TMX and WGX, however, focus on a member's relationship to his or her team as a whole. CWX relationships at the more traditional dyadic level have not been studied nor has the empirical relationship between LMX and CWX. The purpose of the present study was to extend our understanding of the role of CWX in leadership. We examined relationships within coworker dyads to see if the quality of CWX was related to the LMX relationships these coworkers were experiencing. This was our first chance to see how LMX might affect exchanges in other parts of the work group. We also examined the relationship of CWX to employees' work attitudes, after controlling for LMX's influence on those attitudes. CWXs focus on a different referent than leader exchanges, and we expected them to have unique effects on work attitudes.

Theory and Hypotheses

LMX theory (Graen & Uhl-Bien, 1995) holds that leaders develop unique relationships with their various subordinates. Relationships that are of low quality involve exchanges basic to the employment contract. High-quality LMX relationships involve mutual exchanges that go beyond those fundamental to the employment contract. They are characterized by mutual respect and trust and afford greater autonomy and assistance for the member in return for enhanced commitment and loyalty for the leader (Yukl & Van Fleet, 1990). In that body of work, LMX has been portrayed as one type of exchange that is part of a larger network of exchanges, including CWX (Graen & Uhl-Bien, 1995).

In the current study, we sought to understand CWX relationships and to examine if they are related to LMX relationships. We conceptualized CWX as a dyadic process and measured the quality of CWX along similar dimensions as used by LMX. We took this perspective for several reasons. First, the quality of LMX relationships is traditionally viewed in terms of mutual respect, trust, and obligation between leader and member (Graen & Uhl-Bien, 1995). Group theory suggests that these dimensions also are important in coworkers' relationships. Group members provide each other with social support and feelings of personal worth (Sherif & Sherif, 1964). Similarly, in groups that are cohesive, stable, and effective, members have well-developed exchanges leading to feelings of loyalty and trust (Jacobs, 1970). If CWX is related to LMX, that effect may be most evident when similar dimensions of exchange are assessed. As Foa and Foa (1980) pointed out, exchanges tend to be developed within similar classes of resources. People do not exchange money for trust. Beliefs about respect, trust, and loyalty within CWX relationships are likely to be related to those same issues in LMX relationships. As Uhl-Bien et al. (2000) suggested,

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respect, trust, and obligation are key components of quality relationships at all levels—LMX, CWX, TMX, network exchange, and organization–member exchange. Thus, we conceptualized a high-quality CWX relationship as one between coworkers, who report to the same supervisor, that is characterized by mutual respect, trust, and obligation.

The question that we investigated was whether LMX relations have any effect on exchange quality between coworkers. Research on LMX describes leadership relationships as part of a larger network of relationships and suggests that exchanges in one part of that network may affect relationships in other parts (Graen & Uhl-Bien, 1995). Blau (1964) acknowledged that larger relational contexts surround and affect all dyadic exchanges. Similarly, leadership is argued to involve reciprocal influence processes among multiple individuals at different levels (Yukl & Van Fleet, 1990; Sparrowe & Liden, 1997). Sparrowe and Liden (1997), in linking LMX to social network analysis, argued that LMX quality influences the development of subordinates' relationships with others. Adopting this perspective, we believe that LMX quality between a leader and his or her subordinates has a bearing on the relationships forged between those subordinates. Heider's (1958) balance theory provides a useful lens through which multiple dyadic exchanges can be viewed.

Heider's (1958) theory of balance proposes that if Person A and Person B "like" or have "positive sentiment" toward one another, and both share a positive (or negative) sentiment about Person C, they experience "balance." Balance also results when Person A "dislikes" Person B and one has a positive assessment of Person C while the other has a negative assessment. All other configurations are considered unbalanced. Because unbalanced configurations provoke discomfort and tension, Heider suggested that individuals will adapt to achieve balance. As Sparrowe and Liden (1997) succinctly stated "Two close friends of an individual will themselves become friends" (p. 533). McPherson and Smith-Lovin (1987) observed in relationships a homophily bias; individuals who were similar tended to form closer relationships than did dissimilar individuals. Coworkers who both experience high- or both experience low-quality LMX relationships may well perceive themselves to be similar to each other and thus form closer relationships.

It is a small step to argue that such dynamics may be operating in LMX and CWX relationships. If a leader has high-quality LMX relationships with Employee A and Employee B, then balance dynamics suggest that Employee A will develop a high-quality CWX relationship with Employee B (see Figure 1). Similarly, if a leader has low-quality LMX relationships with both Employee A and Employee B, then A and B are likely to experience a high-quality CWX relationship with one another. Finally, when a leader experiences a high-quality LMX relationship with one coworker (A or B) but a low-quality LMX relationship with the other, balance dynamics would predict a weak CWX relationship between A and B. Thus, within a triad of relationships, as depicted in Figure 1, a positive CWX relationship between Employees A and B should be predicted by the similarity of A's and B's LMX relationships.

> *Hypothesis 1:* The more similar the LMX relationships between two coworkers and their leader, the more positive the CWX relationship between those workers is likely to be.

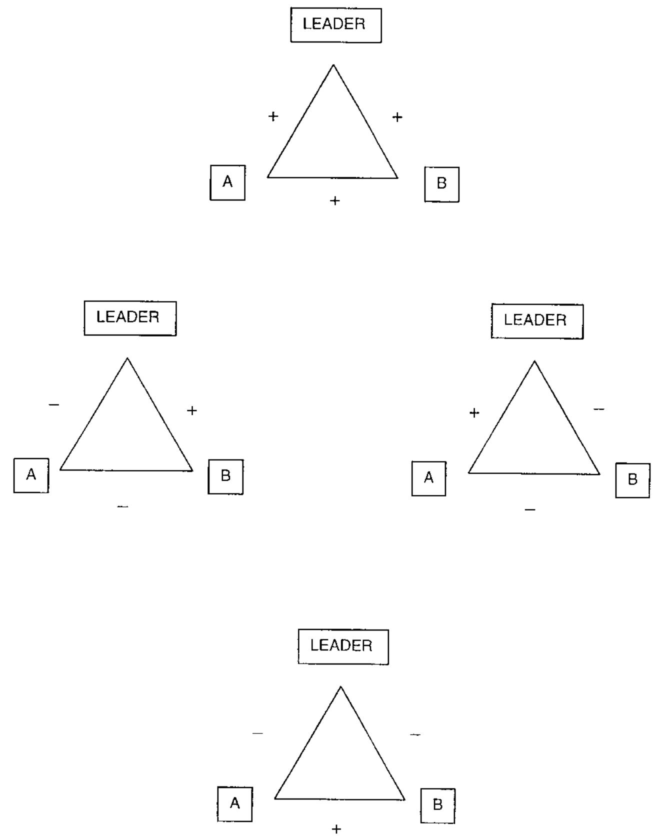

Figure 1. Balanced LMX–CWX triads (LMX = leader–member exchange; CWX = coworker exchange). "A" and "B" refer to different coworkers. The + sign refers to positive relationships and the − sign, to negative ones.

If employees experience CWX and LMX relationships of varying quality, the question arises as to how these different types of exchanges affect employees' work attitudes. The literature provides evidence that LMX influences work attitudes. Liden, Sparrowe, and Wayne (1997) stated that "consistent support has been found for a positive association between LMX and organizational commitment," (p. 67) citing eight such studies. They identified seven studies that have found a positive association between LMX and job satisfaction. Past research on small groups and teams also makes it clear that relationships among teammembers can affect how those members feel about the team and their jobs (Campion, Medsker, & Higgs, 1993; S. G. Cohen & Bailey, 1997; Goodman, Ravlin, & Schminke, 1987). Thus, we expected CWX to affect work attitudes in much the same way LMX does. Given that CWX assesses a set of relationships that is distinct from LMX, however, we expected that CWX would be associated with work attitudes independent of any LMX effects.

The effect of CWX on work attitudes is complicated, however, by the fact that CWX involves exchanges with multiple coworkers. Put another way, a coworker experiences a unique CWX relationship with each member of his or her work group. Hence, we might gain insights into the impact of CWX on attitudes by examining the composition of exchange experiences an employee is having. Yukl (1989) suggested that when a group includes an in-group that is sharply differentiated from the out-group, out-group members may develop feelings of resentment. In such divided groups, members will experience relationships that are diverse in quality. There is reason to believe that such diversity will negatively affect work attitudes. Previous work on group composition has focused mainly on demographic characteristics of group members, such as age, gender, and education (S. G. Cohen & Bailey, 1997). Yet groups can be diverse with respect to nonobservable characteristics, such as values and personality traits, as well. Milliken and Martins (1996) spoke to the difficulties encountered in diverse groups, whether the groups are diverse with respect to nonobservable characteristics or readily detectable ones: "Underlying differences in the schemas, or the conscious and unconscious preconceptions and beliefs, that organize people's thinking can create serious coordination difficulties for groups" (p. 404). Smith et al. (1994) found that diversity can reduce communications and social integration within the group. Hackman (1990) suggested that as a group member has to deal with greater diversity in his or her relationships, such diversity can create tensions within the group. Thus, we hypothesized the following:

> *Hypothesis 2:* Controlling for LMX, employees who report a greater diversity of CWX relationships will have less positive work attitudes.

Work on group composition, however, suggests that an individual within the group may be affected both by the diversity of experiences and by the overall quality of relationships experienced within the group—good or bad (Hackman, 1990). Dunegan, Tierney, and Duchon (1992) found a positive relationship between WGX quality and perceptions of climate. Seers (1989) found a positive association between TMX quality and work attitudes. We expected that a similar pattern might exist for CWX. Higher average levels of CWX should be associated with more positive work attitudes. Averaging CWX scores, however, presents a conceptual problem. To illustrate, a worker with two very high-quality CWX relationships and two very low-quality CWX relationships would have the same average CWX score as would a worker with four mediocre CWX relationships. This example suggests that the average quality of a worker's CWX relationships may be better understood by simultaneously considering the worker's average level of CWX quality and the variation in those CWX scores. Workers who report higher average levels of CWX with little variance in those relationships are experiencing a uniformly better group experience. Such an experience might predict higher levels of satisfaction and commitment.

> *Hypothesis 3:* Controlling for LMX, work attitudes will be more positive when average levels of CWX are higher and diversity in CWX relationships is low.

Method

Sample and Procedure

Data were collected from employees at a Chicago-area engineering company and a health services facility located in eastern Iowa. The sample for this research was composed of 109 employees in 21 work groups. Work groups consisted of three to nine individuals reporting to a particular manager. Each employee was asked to complete a questionnaire that included measures of CWX, LMX, job satisfaction, and organizational commitment. Participants were informed in advance of the nature of the study and were assured that responses would be confidential, that responses would be used for research purposes only, and that participation was completely voluntary.

Sixty-nine employees returned surveys, a 63.3% response rate, of which 67 were usable. Each employee was asked to rate the quality of the relationship he or she had with his or her supervisor, resulting in 67 LMX ratings. Each employee also rated the quality of relationship he or she had with every one of his or her coworkers ($M = 4.82$ coworkers), resulting in 323 CWX ratings. A dyad was created where we had complete information for two employees who had rated one another. Once paired, a total of 110 dyads with complete LMX, CWX, and work attitude data resulted. Ninety-five percent of the original work groups that participated in the study were represented in this final sample.

Of the respondents, 43.3% were male. The average age of the respondents was 36.7 years ($SD = 10.05$). On average, respondents had worked for their employer for 4.24 years ($SD = 4.30$) and had reported to their supervisors for 2.74 years ($SD = 3.26$). One respondent had not completed high school, 38.8% were high school graduates, 7.5% held associate or technical degrees, 37.3% were college graduates, and 14.9% had master's degrees.

Measures

LMX. The LMX7 measure was used to assess LMX. This 7-item scale appropriately captures the three important dimensions of LMX, namely trust, respect, and obligation (Graen & Uhl-Bien, 1995). LMX scores were based on 5-point Likert scales and demonstrated strong internal consistency ($\alpha = 0.93$). Respondents also provided a 1-item rating of each coworker's "working relationship" with the supervisor on a scale ranging from 1 (*one of the worst*) to 5 (*one of the best*).

CWX. CWX quality was measured by adapting the LMX7 measure (Graen & Uhl-Bien, 1995). One item—"How well does your leader recognize your potential?"—was dropped because it did not seem appropriate for coworker relations. The remaining items were rephrased to gauge the respondent's assessment of his or her relationship with a coworker ($\alpha = 0.92$). Principal factor analysis using varimax rotation yielded separate CWX and LMX factors with no cross-loadings, indicating that employees were distinguishing between these two types of relationships.

Job satisfaction and organizational commitment. The 1-item Faces Scale (Kunin, 1955) was used to measure overall job satisfaction. We adapted three questions from Mowday, Steers, and Porter's (1979) 9-item questionnaire to measure organizational commitment ($\alpha = .90$) because of the questionnaire's high reliability and because of company concerns about survey length. The items related to willingness to extend effort for organizational success, similarity between personal and organizational values, and concern about the fate of the organization ($\alpha = 0.82$).

Control variables. We also examined three control variables when testing hypotheses. Company (dummy coded), company tenure, and the number of years the employee had reported to his or her supervisor were considered.

Results

We first addressed the possible concern that respondents might differ in some systematic way from nonrespondents in their exchange relationships. For example, coworkers who had poorer relationships with their supervisors or with other coworkers may have decided to not participate in the study. Because respondents rated all of their coworkers, even those who did not respond to our survey, we had the opportunity to compare respondents and nonrespondents on several ratings. As noted earlier, employees were asked to provide an overall rating of each coworker's "working relationship" with the supervisor. Employees who responded to the survey ($M = 3.72$, $SD = 0.63$) and employees who did not respond ($M = 3.76$, $SD = 0.64$) were not judged to have significantly different relationships with their supervisors, $t(100) = -0.34$. Similarly, we compared the average level of CWX ratings for respondents and nonrespondents that were provided by coworkers in our sample. A comparison of these CWX averages revealed that respondents ($M = 3.57$, $SD = 0.95$) were described as having significantly higher quality CWX relationships than nonrespondents ($M = 3.28$, $SD = 0.88$), $t(319) = 2.47$, $p < .05$. Although these findings represented only respondents' judgments, they did suggest that respondents were seen as having LMX relationships similar to those of nonrespondents but better CWX relationships than those of nonrespondents. The findings presented below need to be considered in light of that difference.

Before testing hypotheses, we also evaluated the convergence in dyad members' CWX ratings. An examination of the level of agreement between coworkers' rating of CWX revealed a median within-group interrater reliability (r_{WG}) of 0.91, suggesting that coworkers viewed the quality of their working relationships similarly (James, Demaree, & Wold, 1984). Given this level of coworker agreement, CWX ratings were averaged to create a single CWX score for each dyad. This score was used in tests of Hypothesis 1. Descriptive statistics for this, and other variables, are shown in Table 1.

Hypothesis 1 proposed that coworkers should have a higher quality CWX relationship when their respective LMX relationships are similar. Because the quality of CWX and the *similarity* of those coworkers' LMX relationships existed only at the dyad level of analysis, we used coworker dyads in testing the hypothesis. It should be noted that even though CWX and the similarity of those coworkers' LMX scores were uniquely defined for each dyad, employees appeared in more than one coworker dyad (more is said about this below). Given this perspective, two different approaches were taken to represent LMX similarity and test its relationship to CWX. One approach used moderated regression and tested the relationship between CWX and an LMX interaction term. The second approach calculated an r_{WG} (James, Demaree, & Wold, 1984) score as an index of LMX similarity within a dyad and used that r_{WG} score to predict CWX.

Before testing Hypothesis 1, we examined the relationships between CWX and the control variables. None of the control variables had significant relationships to CWX, so they were dropped from consideration (see Table 1). We first tested Hypothesis 1 by regressing dyad CWX on the LMX scores for each coworker plus the interaction of the two LMX scores. The LMX interaction term would provide the critical test of Hypothesis 1. If similarity of LMX scores was related to higher quality CWX, then we should find a significant interaction. Moreover, that interaction should take a specific form, where CWX scores were higher when both LMX scores were high or both LMX scores were low.

The overall regression model was significant, $F(3, 106) = 6.54$, $p < .001$, adjusted $R^2 = .13$, as was the interaction term (standardized coefficient $= 1.42$, $p < .04$), suggesting support for Hypothesis 1 (see Table 2). The interaction was interpreted by following Cohen and Cohen's (1983) recommendation to plot the slopes of the interaction. This analysis revealed that when Coworker A's LMX rating was above average (1 *SD* above the mean), higher levels of Coworker B's LMX were associated with higher quality CWX relationships (slope $= 0.27$). When LMX for Coworker A was low (1 *SD* below the mean), however, higher Coworker B LMX ratings were negatively related to CWX quality (slope $= -0.09$). (See Figure 2.) This test provided clear support for Hypothesis 1.

Although these results supported the hypothesis, we were concerned about the dependency that existed within the data set because of employees being in more than one coworker dyad. This concern led us to examine the residual histogram, the normal probability plot of residuals, residual scores within each rater, and the plot of residuals versus the predicted values for our regression model (Neter, Kunter, Nachtsheim, & Waserman, 1996). These plots suggested that the regression model was appropriate and the test of the hypothesis was not biased. In addition, we tested the hypothesis using a general method of moments approach to obtain efficient estimates of our parameters, even if the variance of errors was potentially not constant across observations (i.e., heteroscedastic errors; Gallant, 1987). Again, the critical interaction term was significant (coefficient $= 0.20$, $p < .04$). These follow-up analyses underscored the support we found for Hypothesis 1.

Table 1
Descriptive Statistics and Correlations

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Company	0.41	0.49	—												
2. B tenure	4.90	4.21	−.08	—											
3. B years with supervisor	2.85	2.60	.13	.64	—										
4. A tenure	4.15	3.89	.05	.24	.30	—									
5. A years with supervisor	2.66	2.59	.15	.21	.33	.77	—								
6. A LMX	3.61	0.90	−.27	.08	−.02	.01	−.09	—							
7. A CWX	3.56	1.01	.18	.10	.19	−.12	−.13	.34	—						
8. B LMX	3.85	0.78	−.34	.14	.14	.15	.14	.20	.05	—					
9. B CWX	3.59	0.87	−.02	−.12	−.14	.13	.10	.14	.18	.27	—				
10. CWX	3.57	0.73	.12	−.01	.05	−.01	−.03	.32	.81	.19	.73	—			
11. A satisfaction	3.83	0.88	.27	.02	.20	−.02	−.10	.28	.24	.07	.07	.21	—		
12. B satisfaction	3.91	0.91	−.02	.13	.01	.00	−.01	−.00	.03	.50	.37	.24	.14	—	
13. A commitment	5.85	1.24	.19	.04	.16	.04	−.13	.22	.17	.03	−.06	.08	.49	.11	—
14. B commitment	6.18	0.78	.24	−.15	−.13	−.02	.09	−.11	−.04	.35	.31	.16	.20	.55	.06

Note. Correlation coefficients $> .19$ are statistically significant ($p < .05$). The "A" and "B" prefixes denote the different coworkers in a dyad. LMX = leader–member exchange; CWX = coworker exchange.

Table 2
Summary of Ordinary Least Squares (OLS) Regression Analysis for Variables Predicting Coworker Exchange (CWX)

Variable	CWX
A LMX	−0.72
B LMX	−0.67†
A LMX × B LMX	1.42*
R^2	0.16
$F(3, 106)$	6.54***

Note. Entries are standardized regression coefficients. $N = 110$. LMX = leader–member exchange. The prefixes "A" and "B" refer to different coworkers in a dyad.
† $p < .10$. * $p < .05$. *** $p < .001$.

We also tested Hypothesis 1 using a direct measure of similarity between the two coworkers' ratings of their LMX relationships. In James et al.'s (1984) discussion of r_{WG} as a measure of interrater agreement, it is clear that when ratings of a single target by multiple judges are more similar (i.e., scores fall close to each other on the rating scale), the r_{WG} coefficient for those ratings is higher, indicating agreement among judges. Thus, we believed that r_{WG} could be generalized to our setting and used as index of the extent to which the two employees' LMX ratings were similar, that is, fell close to each other on the rating scale. We calculated an r_{WG} score for the LMX ratings given by each employee in a dyad to represent the extent to which their ratings of LMX were similar or agreed. This r_{WG} score was then correlated with CWX for each dyad. This test of the hypothesis also provided clear support for the hypothesis that coworker dyads have a better quality of exchange when those coworkers have more similar relationships with their supervisor ($r = 0.24$, $p < .02$).

In considering the results, we became concerned that demographic similarity between two coworkers might be confounded with the CWX and LMX findings. To ensure that the relationship between CWX quality and employee LMX was not simply an artifact of demographic similarity between dyad members, we conducted one further test. Age, tenure with company, educational level, and gender were coded for demographic similarity using a commonly accepted technique (see Tsui & O'Reilly, 1989). Then, CWX was regressed on the similarity measures to see if demographic similarity could be affecting CWX. The overall regression model was not significant (F value = 0.39), and none of the measures of demographic similarity were predictive of CWX. Thus, it appeared that demographic similarity, as measured here, was not an alternative explanation for our findings.

Hypothesis 2 predicted that employees who report a greater diversity of CWX relationships would have less positive work attitudes, after controlling for LMX's influence. Before conducting the regression analysis, we examined the relationships between work attitudes and the control variables. Company and years with the supervisor did show some significant relationships to job satisfaction or organizational commitment. Therefore, these variables were included as controls in tests of Hypothesis 2.

A coefficient of variation for CWX ratings was calculated for each employee who provided multiple CWX ratings ($n = 66$). A higher coefficient of variation for an employee represented greater diversity in his or her ratings of CWX relationships. We then tested Hypothesis 2 with two separate regression analyses, one for organizational commitment as the dependent variable and one for job satisfaction. In each case, company, years reporting to the supervisor, and LMX were entered into the equation first, followed by CWX variance. As can be seen in Table 3, support for Hypothesis 2 was evident for organizational commitment but not for job satisfaction. After controlling for the effects of LMX, greater diversity in CWX relationships was related to lower levels of organizational commitment.

Hypothesis 3 proposed that work attitudes should be more positive when CWX is higher in quality and lower in diversity. We used moderated regression to test these relationships. The predictors of work attitudes were company, years with supervisor, LMX, the mean of CWX ratings provided by each employee, the coefficient of variation for CWX ratings, and the interaction term for mean CWX and CWX variance. The critical test of the hypothesis was the interaction term. The hypothesis was not supported for job satisfaction (standardized coefficient for the interaction term = −1.00, p = *ns*) or for organizational commitment (standardized coefficient for the interaction term = −0.21, p = *ns*).

Discussion

The relationship between CWX and LMX exchanges was found to support current thinking about how different types of exchanges might influence each other. As hypothesized, the interaction of

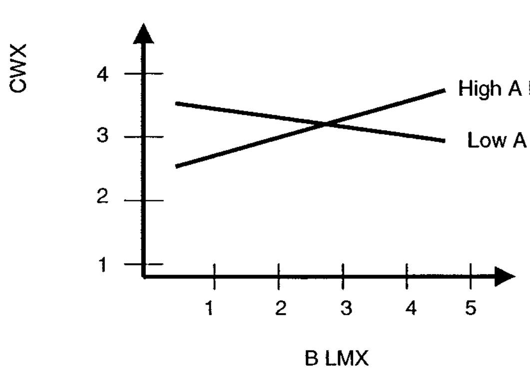

Figure 2. Interaction of A LMX and B LMX. LMX = leader–member exchange; CWX = coworker exchange. "A" and "B" refer to different coworkers.

Table 3
Summary of Ordinary Least Squares (OLS) Regression for Variables Predicting Work Attitudes

Variable	Organizational commitment	Job satisfaction
Company	0.22†	0.30*
Years reporting to supervisor	−0.16	−0.13
LMX	0.32**	0.38**
CWX variance	−0.31**	0.17
R^2	0.23	0.19
$F(4, 61)$	4.50**	3.60*

Note. Entries are standardized regression coefficients. $N = 66$. LMX = leader–member exchange; CWX = coworker exchange.
† $p < .10$. * $p < .05$. ** $p < .01$.

LMX scores within a dyad of coworkers was predictive of CWX between those coworkers. Specifically, when Employees A and B both had high-quality LMX relationships with their supervisor or both had low-quality LMX relationships, they experienced a high-quality CWX relationship with one another. This finding is consistent with the argument that within a triad of dyadic relationships there will be a tendency to move toward balance.

The causal direction of this phenomenon is unresolved, however. A number of possibilities exist that are all consistent with balance dynamics. It may be that employees with similar quality LMX relationships seek to associate with others of like kind. Thus, both in-group and out-group members would form strong relationships amongst themselves. An alternate explanation of this phenomenon is also plausible. Employee A, having strong relationships with the supervisor and Employee B, may persuade the supervisor as to the merits of Employee B, fostering a better LMX relationship between the supervisor and Employee B. Or, in the reverse, Employee A, having a strong LMX relationship with the supervisor and a good CWX relationship with Employee B, may serve as a persuasive advocate for the supervisor. Essentially, Employee A influences Employee B to interact with or to see the supervisor in a more positive fashion. Meindl (1990) suggested that this process of social contagion can affect how leaders are perceived. Future research needs to replicate these findings and to discern the mechanisms by which balance is fostered.

The finding that the similarity between two workers' LMX scores is associated with positive CWX raises some interesting possibilities for theoretical extensions of LMX work. LMX research has not made clear what balance of high- and low-quality LMX relationships is optimal for a leader (Graen & Uhl-Bien, 1995; Sparrowe & Liden, 1997; Yukl & Van Fleet, 1990). Although supervisors might best use their limited time by fostering high-quality relationships with only a handful of work group members, there is also an argument to be made for each member of a team being afforded equal time and similar opportunities to form a high-quality LMX relationship. In fact, if LMX relationships drive CWX relationship quality, it would seem advisable for supervisors to develop high-quality LMX relationships with all subordinates. In doing so, leaders would foster positive CWX relationships that could favorably affect organizational commitment and might enhance team development, group functioning, and cohesiveness. Failure to do so could result in subgroup polarity, pose challenges in managing tension among coworkers, and reduce workers' organizational commitment.

If the causal direction is such that Worker A's high-quality LMX and CWX relationships influence Worker B's LMX quality, it may be that CWX can serve as a multiplier of LMX effects. By leaders fostering strong CWX relationships, positive LMX relationships could spread throughout the work group. Given LMX's persistent positive relationship to outcomes like satisfaction and commitment, such a multiplier effect would deliver considerable value to a leader. However, negative LMX relationships would be equally contagious among coworkers with strong CWX relationships. In that case, leaders could face formidable and cohesive resistance from members of their work group. Thus, one can see a potential intersection of LMX theory with other theoretical streams, such as team development, socialization processes, and mentoring. By examining the integration of LMX with these other processes, we may better understand the impact of exchanges beyond the leader–member relationship.

This study also investigated the relationship between CWX and work attitudes. As predicted, greater diversity in an employee's CWX relationships was found to be negatively related to his or her organizational commitment. This finding suggests that there is some burden involved in managing diverse relationships with coworkers. This pattern could only have been found when studying exchanges in a network of relationships, and it points to the value of extending research on exchanges beyond the dyadic LMX relationship. In contrast, job satisfaction was unrelated to diversity in relationship quality. Why is it that workers remain satisfied with their jobs in spite of the range of quality in their coworker relations while their commitment level suffers? Perhaps exchange diversity, which measures variation in respect, loyalty, and trust, does not affect the pleasure found in the tasks performed at work. Alternately, even though our single item measure of job satisfaction is considered to be a reasonable overall evaluative judgment of job satisfaction (Brief & Roberson, 1989; Fisher, 2000), the use of a single-item measure may have restricted our findings. It also could be the case that our small sample size limited our power in some analyses.

Our study of the aggregate effects of multiple exchanges included an assessment of overall CWX quality level and employee work attitudes. Because each employee has relationships with a number of different coworkers, an average CWX score and CWX variance were used in this analysis. Unlike previous findings that link LMX to positive work attitudes, overall CWX did not appear to be related to positive work attitudes. One interpretation is that LMX represents a key, and unavoidable, workplace relationship that significantly influences an employee's work life. CWX relationships may not be as central to the work experience. Perhaps CWX would be a more powerful predictor of work attitudes if we could identify "significant" coworkers in the network or coworkers that represent critical dependencies for the employee. For example, a worker may experience generally good CWX relationships with all coworkers except for one. If that single low-quality CWX relationship stemmed from a circumstance, such as sexual harassment, whereas one's mean CWX might be above average and CWX variance might be low, it would be entirely possible for work attitudes to be quite negative for that worker.

Given these preliminary findings, future research needs to examine the construct validity of CWX by exploring its relationship to other measures of coworker trust, loyalty, and mutual respect. That research also needs to explore just how LMX and CWX play on one another and ultimately link to work attitudes. Innovative designs that reduce the opportunity for common method bias are also needed. In this work, relationships between exchange measures and work attitudes are contaminated by same-source bias. Nevertheless, several factors reassure us that the findings represent more than method bias. First, CWX showed significant convergence between independent parties in a coworker dyad. Thus, we have confidence that CWX is not just the cognitive construction of a single person but is an observable relationship between coworkers. Also, the differential predictions of organizational commitment and job satisfaction by CWX argue that more than method variance is at play. Finally, the finding of a relationship between the variance of CWX exchanges and commitment does not seem easily explained by method bias. Nevertheless, this measurement challenge needs more attention.

In conclusion, we are encouraged by the findings presented here. It should be noted, however, that the data were collected at two firms and our sample of employees had higher CWX ratings than employees who did not participate. Thus, we cannot be certain about the generalizability of these results. Nevertheless, CWX appears to be a viable and interesting addition to the exchange framework. Moreover, these findings reinforce our interest in pursuing a better understanding of networks of exchanges and the leader's role in those networks.

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